



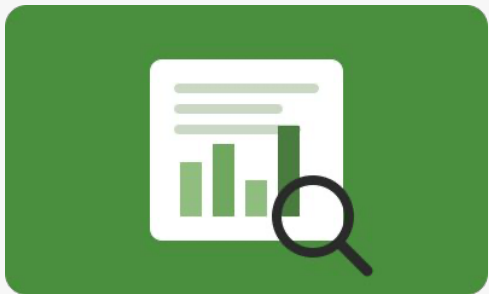
CAHSI Data Analytics Challenge 2026

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Three Ways to Use Data

A running example: a video-streaming app deciding what to show you



DESCRIPTIVE · HINDSIGHT

What happened?

Look at data from the past to summarize and understand it.

Example

"What styles of shows do you watch the most?"



PREDICTIVE · INSIGHT

What will happen?

Use past patterns to make an educated guess about the future.

Example

"What show are you likely to want next?"



PRESCRIPTIVE · FORESIGHT

What should we do?

Turn that guess into a concrete recommendation or action.

Example

"Which show should we recommend right now?"

Predictive Analytics: Examples

Using past data to make an educated guess about what happens next



Healthcare

Will this patient recover well after surgery?



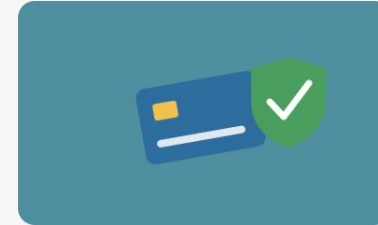
Travel

Will airfare prices drop before I need to book?



Lending

Will this credit applicant default on a loan?



Banking

Is this transaction normal, or likely fraud?

Machine Learning in Cybersecurity

Teaching a computer to recognize patterns that signal danger



Fraud Detection

Flags suspicious purchases before they go through.



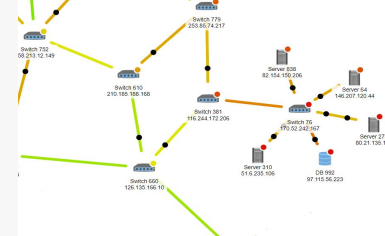
Phishing Detection

Spots fake emails designed to steal your info.



Malware Analysis

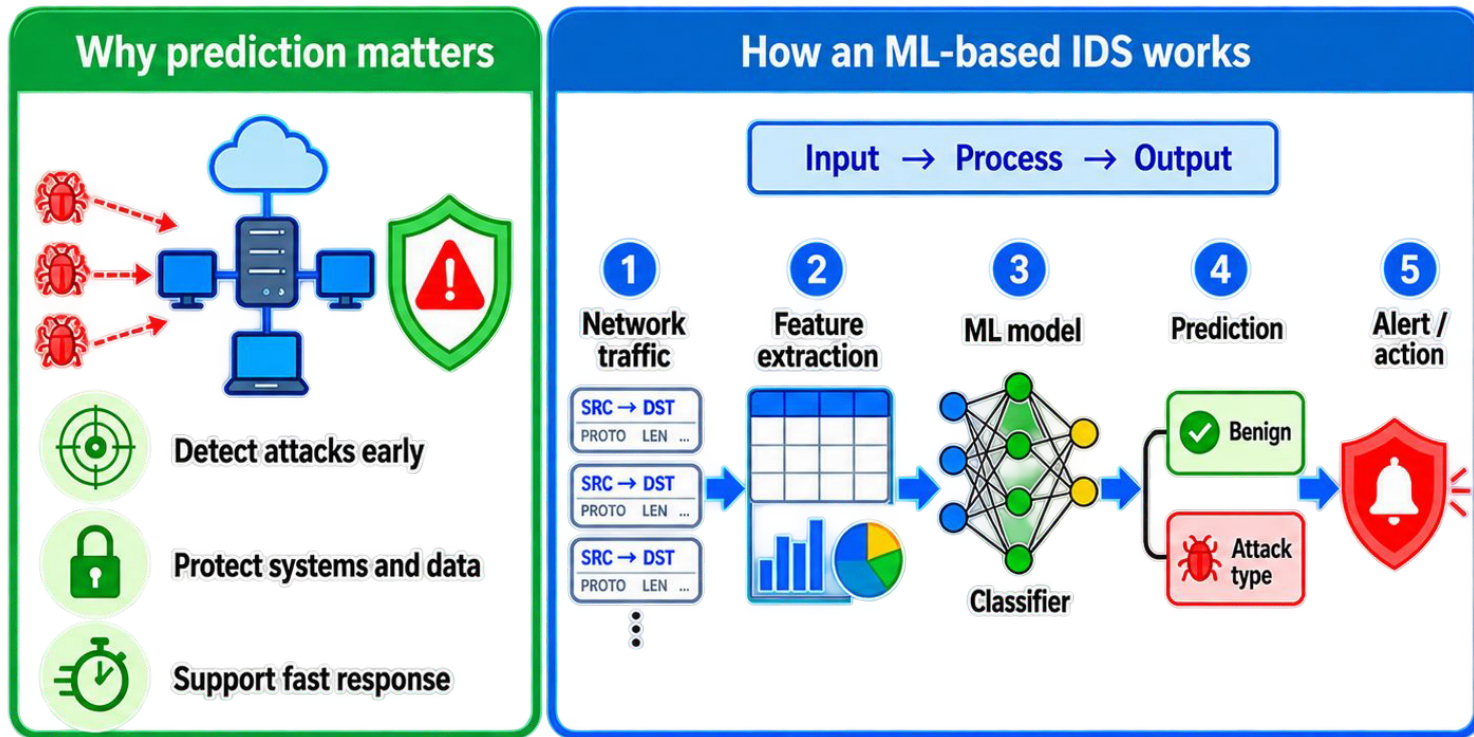
Recognizes software built to damage or steal data.



Traffic Classification

Sorts network traffic to catch unusual connections.

This Session: Predicting Threads in Network Traffic



This Session: Predicting Threats in Network Traffic

- Can a machine learning model distinguish benign network traffic from different types of cyber-attacks?
- **Target variable:** the traffic class
 - Normal (Benign).
 - Several different attack types.
- **Explanatory variables:** Features that describe each network flow and help predict its traffic class. For example:
 - Flow Duration – how long the network communication lasts.
 - Total Packets – how many packets are exchanged.
 - Flow Bytes/s – how much data is transferred per second.

Sample of Dataset D1

Protocol: The communication protocol used by the network flow

Flow Bytes/s & Flow Packets/s: Measure how much data and how many packets are transmitted per second during a network flow.

Flow Duration: how long the network communication lasts

Total Forward & Backward Packets: Count the number of packets sent in each direction during a network flow.

Mean Packet Length: The average size of packets transmitted during a network flow.

Class Label

Features →

Samples {

Protocol	Flow Duration (microseconds)	Total Forward Packets	Total Backward Packets	...	Flow Bytes per Second	Flow Packets per Second	Mean Packet Length	Label
UDP	51,236	4	4	...	19,517.5	156.1	125	benign
UDP	53,353	4	4	...	18,743.1	149.9	125	benign
TCP	1	2	0	...	1.2e+08	2e+06	60	syn
TCP	2	2	2	...	1.2e+08	2e+06	60	syn
TCP	163,980	8	6	...	78,503.5	85.4	919.5	dos hulk
TCP	163,983	8	6	...	78,502.0	85.4	919.5	dos hulk
...	...	...	...	...	...	...	...	...

This sample shows **7 of the 92 features** in the dataset. The full D1 contains **1,704,464 samples** and **36 unique classes**.

Similarities in D1

- Similar samples often have similar target values.
- Patterns in historical data can help us predict new, unseen samples.
- Machine learning helps us capture these patterns automatically.
- How can we measure similarity or learn these patterns from the data?

Features →		Protocol	Flow Duration (microseconds)	Total Forward Packets	Total Backward Packets	...	Flow Bytes per Second	Flow Packets per Second	Mean Packet Length	Label
Samples {		UDP	51,236	4	4	...	19,517.5	156.1	125	benign
		UDP	53,353	4	4	...	18,743.1	149.9	125	benign
		TCP	1	2	0	...	1.2e+08	2e+06	60	syn
		TCP	2	2	2	...	1.2e+08	2e+06	60	syn
		TCP	163,980	8	6	...	78,503.5	85.4	919.5	dos hulk
		TCP	163,983	8	6	...	78,502.0	85.4	919.5	dos hulk
		...	...	...	...	...	...	...	...	...

This sample shows **7 of the 92 features** in the dataset. The full D1 contains **1,704,464 samples** and **36 unique classes**.

Label Mapping

id	class
0	analysis
1	backdoor
2	benign
3	bot
4	ddos
5	dos
6	dos goldeneye
7	dos hulk
8	dos slowhttptest
9	dos slowloris
10	drdos_dns
11	drdos_ldap

id	class
12	drdos_mssql
13	drdos_netbios
14	drdos_ntp3
15	drdos_snmp
16	drdos_ssdp
17	drdos_udp
18	exploits
19	ftp-patator
20	fuzzers
21	generic
22	infiltration
23	mssql

id	class
24	portscan
25	reconnaissance
26	shellcode
27	ssh-patator
28	syn
29	tftp
30	udp-lag
31	web attack – brute force
32	web attack – sql injection
33	web attack – xss
34	webddos
35	worms

Sample of Dataset D2

- How are these samples similar to the rows in D1?
- Can we measure their similarities?
- Similar rows are more likely to belong to the same Label.

Missing the label column from D2.

Task: Predict 1 of the 35 possible classes for each sample.



Features →		Protocol	Flow Duration (microseconds)	Total Forward Packets	Total Backward Packets	...	Flow Bytes per Second	Flow Packets per Second	Mean Packet Length	Label
Samples {		TCP	3	2	2	...	8e+07	1.33e+06	60	?
		TCP	164,999	8	6	...	78,018.7	84.8	919.5	?
		UDP	47,196	4	4	...	21,188.2	169.5	125	?
		TCP	1	2	0	...	1.2e+08	2e+06	60	?
		UDP	64,204	4	4	...	15,575.4	124.6	125	?
		TCP	162,993	8	6	...	78,978.9	85.9	919.5	?
		...	...	...	...	...	...	...	...	...

This sample shows **7 of the 91 features** in the dataset. The full D2 dataset contains **312,107 samples**.

The Task: Classify Network Traffic

- You will be given:
 - D1: A large dataset **with traffic labels** (1.22 GB).
 - D2: A large dataset **without traffic labels** (235.9 MB).
- Use D1 to learn the **relationship between the explanatory variables (features) and the target variable (traffic class)**.
- Then, use your trained model to predict the traffic class for each row in D2.

Tools:

- Any machine learning or data analytics tools.
- Any generative AI tools.

Logistics

- Competition duration
 - Date: September 18, 2026
 - Starting: 8:00 AM, Mountain Daylight Time (MDT)
 - Ending: 11:00 AM, MDT

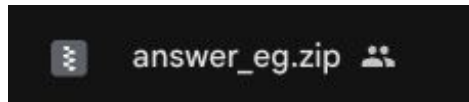
Get the Data and Submit Your Work - Local Disk

- Get the data here: <https://tinyurl.com/gmis2026-data>.

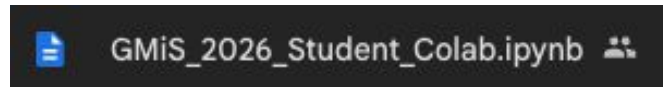
Download the data:



Download the sample submission:



Sample code for Google CoLab:

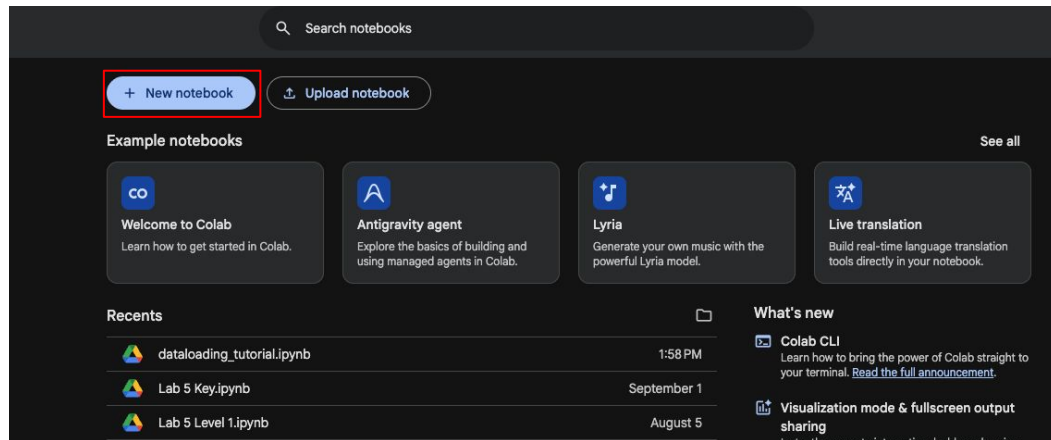


- Submit your work
 - Through the CAHSI Data Analytics Hackathon ([the link is provided in later slides](#)).
 - Detailed instructions will be explained in the following several slides.

Get the Data and Submit Your Work - Colab

You can also download and use the data **directly** in **Google Colab**:

1. Go to <https://colab.research.google.com/>.
2. Open a new notebook.



Get the Data and Submit Your Work - Colab

You can also download and use the data **directly** in **Google Colab**:

1. Go to <https://colab.research.google.com/>.
2. Open a new notebook.
3. Run code to download the data.

```
 # Downloads the training dataset 'D1.pkl' to drive,  
# and the testing dataset 'D2.pkl' to drive.  
  
import requests  
  
D1_link = "https://tinyurl.com/D1-data"  
D2_link = "https://tinyurl.com/D2-dataset"  
  
response_d1 = requests.head(D1_link, allow_redirects=True)  
response_d2 = requests.head(D2_link, allow_redirects=True)  
  
real_url_d1 = response_d1.url  
real_url_d2 = response_d2.url  
  
!gdown --fuzzy $real_url_d1  
!gdown --fuzzy $real_url_d2
```

Get the Data and Submit Your Work - Colab

You can also download and use the data **directly** in **Google Colab**:

1. Go to <https://colab.research.google.com/>.
2. Open a new notebook.
3. Run code to download the data.
4. Run code to use the data.

```
 # Load data using Pandas

import pandas as pd

df_train = pd.read_pickle("/content/D1.pkl")
df_test = pd.read_pickle("/content/D2.pkl")

print(df_train.head())
```

Get the Data and Submit Your Work - Colab

You can also download and use the data **directly** in **Google Colab**:

1. Go to <https://colab.research.google.com/>.
2. Open a new notebook.
3. Run code to download the data.
4. Run code to use the data.
5. Save and download the predictions into **answer.txt**.
6. Zip answer.txt and upload.



```
1 from google.colab import files
2
3 # --- Simulate ML Predictions (Replace with your actual model predictions) ---
4 # For demonstration, let's create some dummy predictions.
5 # In a real scenario, 'predictions' would be the output of your ML model.
6
7 # You might have numerical predictions in the range [0-35], e.g.:
8 dummy_predictions = [32, 2, 16, 20, 33]
9
10 with open("answer.txt", "w") as f:
11     # Join predictions with newline characters, ensuring no trailing newline
12     f.write("\n".join(map(str, dummy_predictions)))
13
14 # --- Download the answer.txt file ---
15 files.download("answer.txt")
```

Submit Your Work - Prepare Your Answer

1. Look at the sample submission →



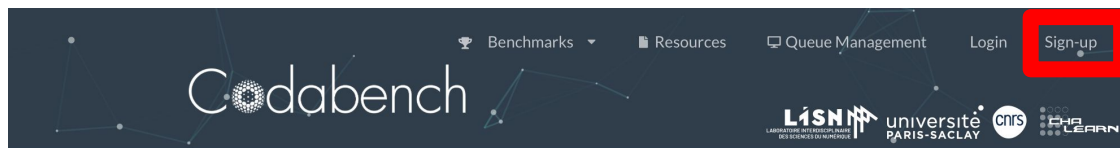
- Uncompress this file, you will get answer.txt, from which you can understand how you will organize your submitted file.
2. Submit this **answer_eg.zip** through the steps shown in the next few slides.

Submission Guidelines

- a. Your answer file **MUST** be named **answer.txt**.
- b. answer.txt **MUST** contain one prediction **per instance of D2.pkl**.
 - Each prediction in answer.txt should have its own row.
- c. You **MUST** **zip your answer file**. You can then rename the zipped file as *answer_team1_attempt2.zip* or *answer_team2_attempt2.zip*, etc.
- d. Submit your **zipped answer file**.

Submit Your Work - Preparation - Sign up

Go to <https://www.codabench.org/accounts/signup/> and create an account for **your team**.



ne to **Codabench**! New here? Check out our [getting started guide](#) to begin your journey with ease!

Join the [Google group](#) to connect with the community!

or small jobs (typically for result submission competitions). For code submission, organizers are expected to provide their own [see instructions](#). The execution time is currently capped at 20 minutes on the default public queue.

Recent Benchmarks

ed Emotion isk 11 - Track A alived emotion(s)	September 16, 2024 782 Participants	Task B- sinhala language  Organized by: <i>ranlpss</i>	September 4, 2025 6 Participants
PS 2025 Competition er ...	July 30, 2025  765 Participants	TaskB-Portuguse  Organized by: <i>ranlpss</i>	September 4, 2025 7 Participants

Sign Up

☐ I accept the [terms and conditions](#).

Sign Up

Already have an account? [Log In](#) instead.

Submit Your Work - Preparation - Sign in

Go to <https://www.codabench.org/accounts/login/> and sign in.

Login

 username or email

 password

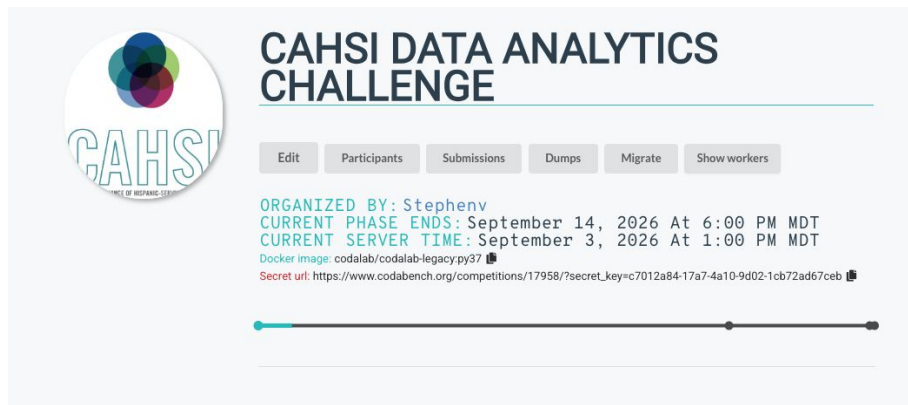
Log In

Don't have an account? [Sign Up](#)

[Forgot your password?](#)

Submit Your Work - After You Are Approved to Participate

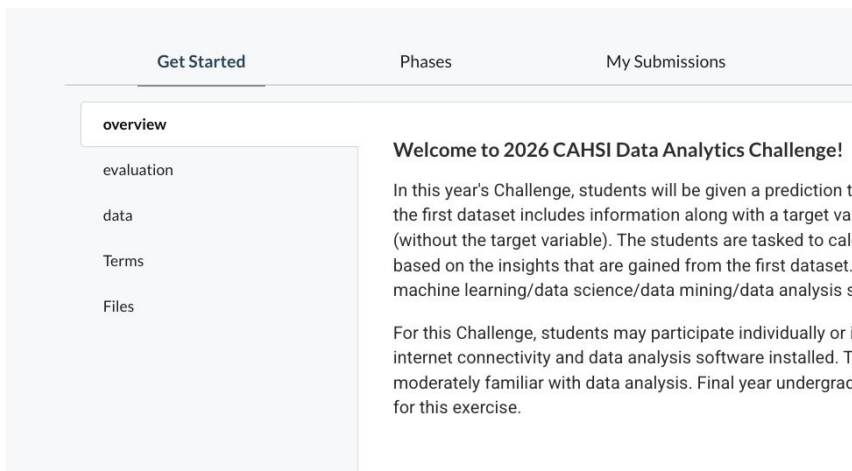
In a browser, go to the competition (the link is provided at a later slide)



CAHSI DATA ANALYTICS CHALLENGE

ORGANIZED BY: [Stephenv](#)
 CURRENT PHASE ENDS: September 14, 2026 At 6:00 PM MDT
 CURRENT SERVER TIME: September 3, 2026 At 1:00 PM MDT
 Docker image: [codalab/codalab-legacy:py37](#)
 Secret url: https://www.codabench.org/competitions/17958/?secret_key=c7012a84-17a7-4a10-9d02-1cb72ad67ceb

You will see an interface similar to this →



Get Started Phases My Submissions

overview

- evaluation
- data
- Terms
- Files

Welcome to 2026 CAHSI Data Analytics Challenge!

In this year's Challenge, students will be given a prediction to the first dataset includes information along with a target variable (without the target variable). The students are tasked to calculate based on the insights that are gained from the first dataset. 1 machine learning/data science/data mining/data analysis sc

For this Challenge, students may participate individually or in internet connectivity and data analysis software installed. They are moderately familiar with data analysis. Final year undergraduate for this exercise.

Submit Your Work - Participate

Click tab “[My Submissions](#)”, you will see an interface showing similar to what follows.

Get Started

Phases

My Submissions

You have not yet registered for this competition.

To participate in this competition, you must accept its specific [terms and conditions](#).

This competition **requires approval** from the competition organizers. After submitting your registration, the organizers will notify them of your request. Your application will remain pending until they approve.

☐ I accept the terms and conditions of the competition.

Register

Accept the terms and the conditions

Click the “[Register](#)” button

Submit Your Work - Participate

After you click your register button, you need to wait for us to approve your registration. You should see a message like this when your [registration is pending](#).

[Get Started](#)[Phases](#)[My Submissions](#)[Results](#)[Forum](#)[?](#)**Registration Status: Pending**

Your request to participate in this competition is waiting for an approval from the competition organizer.

Submit Your Work - Participate

After 2-3 seconds, refresh your browser. You should see an interface similar to what is shown below.

The screenshot displays a web interface for managing submissions. At the top, there is a navigation bar with tabs: 'Get Started', 'Phases', 'My Submissions' (which is the active tab), 'Results', and 'Forum'. A help icon (?) is located on the far right of this bar. Below the navigation bar, there are two buttons: 'Test Submission' and 'Competition'. The main content area is divided into two sections. The first section contains two metrics: 'Number of submissions used for the day' and 'Number of total submissions used', both showing '6 out of 999' in green boxes. The second section is titled 'Submission upload' and includes a 'Submit as:' dropdown menu currently set to 'Yourself', a help icon (?), and a large text input field with a paperclip icon for file uploads.

Get Started	Phases	My Submissions	Results	Forum	?
Test Submission Competition					
Number of submissions used for the day 6 out of 999 Number of total submissions used 6 out of 999					
Submission upload Submit as: ? Yourself					

Submit Your Work - Participate

- Click on the paperclip.
- The system will prompt you to upload your answer file. Choose [your answer_teamx_attempty.zip](#).

Submission upload

Submit as: ?

Yourself



Submit Your Work - Participate

Now the dashboard should show your submitted files, submission dates, and status. Your recent submission status should be “Submitting” or “Submitted”.

Submission upload

Submit as: ?

Yourself



▶ Running answer_myteam_attempt1.zip (ID = 368916)

Search...  Status 

ID # ▼	File name	Date	Status	Score	Actions
368916	answer_myteam_attempt1.zip	2025-09-11 18:22	Submitted		

Submit Your Work - Participate

- [Refresh the page](#), your status will be changed to “[Preparing](#)”, “[Running](#)”, or “[Finished](#)”.

ID # ▾	File name	Date	Status	Score	Actions
368919	answer_myteam_attempt3.zip	2025-09-11 18:43	Finished	0.67	 

- Once the status changes to “Finished”, your best result will be automatically uploaded to the leaderboard.

Submit Your Work - Participate - Show Results

Click the “[Results](#)” tab, you should be able to see your best result and your rank.

[Get Started](#)
[Phases](#)
[My Submissions](#)
[Results](#)
[Forum](#)
?

[Test Submission](#)
[Competition](#)

?

Number of submissions used for the day

3 out of 999

Number of total submissions used

9 out of 999

Results				
Task:				Competition Task
#	Participant	Date	ID	Macro-F1
	kasun	2026-09-03 11:04	913950	0.7789

Evaluation Metrics: Macro-F1 Score

- Macro-F1 is an evaluation metric that calculates the unweighted average of the F1-scores for each individual class.
- Reference: <https://tinyurl.com/f1metric>.

Prizes & Team Formation

- Certificates
 - The first three teams will receive certificates from GMiS.
 - The winners will be announced at the University 'Night' on Saturday afternoon (4:30-7:30 PM).
- Team formation
 - With 1-3 people, i.e., you can form a team (1) with only yourself, or (2) with yourself and 1 or 2 extra people.
 - Please create a TEAM name.
 - Submit your team formation through this form (<https://tinyurl.com/GMiS-DAC-Teams>).

Before Start - Go Through the Submission

- Set up a competition account for your team at <https://codabench.org>.
- Register for the competition ([please make sure to get approval before you leave this session](#)).
- Download the datasets and answer_eg.zip and file from <https://tinyurl.com/gmis2026-data>.
- Submit the answer_eg.zip to the competition system.
- Check results.
- Competition link: <https://tinyurl.com/GMiS-DAC2026>.

Submit Your Source Code

- If your team achieves an F1 score ≥ 0.5 , please submit your source code.
- Your README must clearly describe: the software used, the required libraries, and the steps to run your model.
- Compress the following into one ZIP file: (1) your source code, (2) README file, and (3) The answer file that achieved your highest F1 score, name the zip file with the names of the team members.
- Submit the required information through the Google Form:
<https://tinyurl.com/GMiS-DAC-Code-Submission>.
- Submit your ZIP file through the Google Form within 30 minutes after the competition ends. (e.g., the submission window is 8:00AM - 11:30AM, MDT).
- Submissions received more than 30 minutes after the competition ends will not be considered.
- If multiple submissions are received from the same team, only the last submission received within the valid window will be considered.
- Teams that do not submit source code may not be considered for the final ranking.